

# The Knowledge Lifecycle of Large Language Models

A model holds two memories. When they disagree, training usually wins.

Acquire

Store

Retrieve

Update

Forget



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## Start with a question

You paste a document into a model's prompt.

The document contradicts what the model learned.

**Which one wins?**

Most of us assume the prompt wins. That assumption is what this talk tests.

# The field is fragmented

- Five research communities each own one part of how a model handles a fact:
  - retrieval-augmented generation · knowledge editing
  - continual learning · context engineering · agent memory
- Each has its own benchmarks. Each passes them.
- A deployed system can pass all five and still fail, because nothing tests what happens **between** them.

Across 20 representative papers, median coverage is **two** of five stages.

## The framework: five stages

| Stage             | What happens to the fact                  | Studied as                            |
|-------------------|-------------------------------------------|---------------------------------------|
| ■ <b>Acquire</b>  | Training compresses a corpus into weights | Pre-training, fine-tuning             |
| ■ <b>Store</b>    | Weights, an external index, or both       | Parametric memory, vector stores      |
| ■ <b>Retrieve</b> | Attention recalls, or a pipeline fetches  | Retrieval-augmented generation        |
| ■ <b>Update</b>   | The world changes; copies must follow     | Knowledge editing, continual learning |
| ■ <b>Forget</b>   | Removed on purpose, or lost by accident   | Unlearning, catastrophic forgetting   |

The failure in this talk sits between **Retrieve** and **Update**.

## A documented case: Vioxx

- Rofecoxib, sold as Vioxx, was approved in 1999 as a COX-2 inhibitor.
- Withdrawn worldwide in **September 2004** after trials showed raised risk of heart attack and stroke.
- A model trained on pre-2004 text holds “approved”. The correction exists, and can be retrieved.



## The measurement

|                     | Without the notice | With the notice in the prompt |
|---------------------|--------------------|-------------------------------|
| Answers “safe”      | 37.53%             | <b>42.58%</b>                 |
| Answers “withdrawn” | 0.0004%            | 0.0006%                       |
| Most likely word    | “safe”             | “safe”                        |

- The withdrawal notice is **in the prompt**, and confidence in “safe” **rises**.
- The correct answer is left at roughly **six chances in a million**.
- Retrieval succeeded. Resolution failed.

# This is not hallucination



No benchmark for retrieval, editing, or memory tests that second step,  
**because it belongs to none of them.**

# The metric: Lifecycle Desynchronization

$$\hat{\mathcal{D}}_{sync} = -\ln P(y^* \mid q, \theta_{stale}, \mathcal{M}_{now})$$

- The surprisal of the **correct** answer while the corrective document is present.
- Measured in nats:  $n$  nats means the right answer holds probability  $e^{-n}$ .
- Above 9.2 nats, no realistic decoding recovers it.



## Two numbers, not one

|                           | $\mathcal{I}_{ctx}$ low | $\mathcal{I}_{ctx}$ high      |
|---------------------------|-------------------------|-------------------------------|
| $\mathcal{D}_{sync}$ low  | fact never conflicted   | context resolved the conflict |
| $\mathcal{D}_{sync}$ high | resolution failure      | drift: influential but losing |

- $\mathcal{I}_{ctx}$  measures whether the document moved the model **at all**.
- On Vioxx it is **0.033** nats. The document is **inert**.
- That pairing is what separates a broken retriever from a model that ignores what it retrieved.

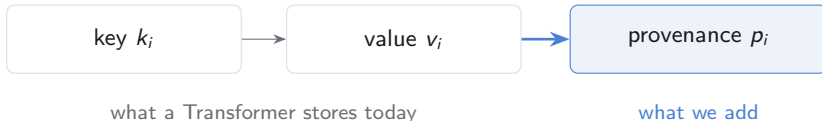
## Three cases, three regimes

| Fact change             | What the model does                 | Regime             |
|-------------------------|-------------------------------------|--------------------|
| Vioxx withdrawn, 2004   | ignores the notice entirely         | resolution failure |
| Elizabeth II died, 2022 | the notice boosts “Queen”           | the exposure trap  |
| Twitter renamed X, 2023 | moves hard, still answers “Twitter” | drift              |

The middle row is the uncomfortable one: **a correct document made the wrong answer stronger.**

# The cause: storage discards provenance

- Training teaches a model **what** is true.
- It never records **when** the fact was learned, or **where** it came from.
- So at inference there is no principled basis for preferring fresh evidence over a confident old memory.



## The proposal: gate stale facts at inference

$$\text{FFN}_{\text{steered}}(x) = \sum_{i=1}^{d_m} S(\Delta\tau_i) f(x \cdot k_i) v_i, \quad S(\Delta\tau_i) = \sigma(\gamma - \beta \cdot \Delta\tau_i)$$

- Fresh facts pass with  $S \approx 1$ ; stale facts are attenuated toward 0.
- **The weights are never modified**, so the specificity bottleneck that limits static editing does not apply.
- Cost: about 1M parameters on a 7B model, under **0.02%**.

Think of it as a volume control on every stored fact, turned down by age.

# What is measured, and what is proposed

## Measured

The metric, on real models,  
deterministically,  
reproducible on a laptop

## Proposed

The architecture,  
with an idealized proof  
under stated assumptions

The Provenance Vector has **not** been trained. It needs a pretraining run, and the paper says so in its limitations rather than leaving you to find out.

# Check it yourself

- Every number in this talk comes from **one deterministic script**. No sampling, identical digits on any machine.
- The model is GPT-2 base, 124M parameters: small, fully open, and free of instruction tuning that would confound the result.
- The demonstration runs GPT-2 **inside your browser**. Nothing you type leaves your laptop.

**Paper and code** [github.com/Amey-Thakur/LLM-KNOWLEDGE-LIFECYCLE](https://github.com/Amey-Thakur/LLM-KNOWLEDGE-LIFECYCLE)

**Live demonstration** [huggingface.co/spaces/ameythakur/llm-knowledge-lifecycle](https://huggingface.co/spaces/ameythakur/llm-knowledge-lifecycle)

## What comes next

- **Does it survive scale? Yes.** Across six open models from 124M to 1.5B, not one answers the Vioxx probe correctly, and inside the GPT-2 family the measurement is not even monotone in size.
- **Does instruction tuning fix it? Not on this probe.** The tuned models resolve the Twitter and Monarch conflicts and still lose Vioxx; the largest answers “unsafe”, which is the right direction and the wrong token.
- **Does reasoning-mode inference help?** Extended test-time compute should, in principle, resolve conflicts the forward pass cannot.
- **Multimodal.** A text prior overriding a contradicting image is the same failure in another modality.

Both answered, and both answers are largely negative.

## In three sentences

- Five separate research fields are five stages of **one lifecycle**, and systems fail at the boundaries nobody benchmarks.
- Those failures are **measurable**: a model with the correction in its prompt still answers wrong, at 12.05 nats.
- The cause is that storage **discards provenance**, and an inference-time gate can address it without touching the weights.



# Thank you

Questions



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[github.com/Amey-Thakur/LLM-KNOWLEDGE-LIFECYCLE](https://github.com/Amey-Thakur/LLM-KNOWLEDGE-LIFECYCLE)

## Backup: anticipated questions

- **Why GPT-2 and not a frontier model?** Small, fully open, no instruction-tuning confound, and it makes the result checkable on a laptop. The claim is that the failure exists and is measurable, not that it is universal.
- **Is this just hallucination?** No. The retriever succeeded, and  $\mathcal{I}_{ctx} = 0.033$  proves the context was present and inert.
- **Does it disappear at scale?** Unknown, and that is the honest answer. The sweep is running; we will report what it shows.
- **Why not just edit the weights?** Editing damages neighboring facts, and degrades as edits accumulate. Gating leaves the weights untouched.
- **Is the proof meaningful if untrained?** It shows the mechanism is structurally sound given accurate provenance. The assumptions are stated, not hidden.